



WooJin Son

MIR with DSP



Contents

What is MIR & DSP?



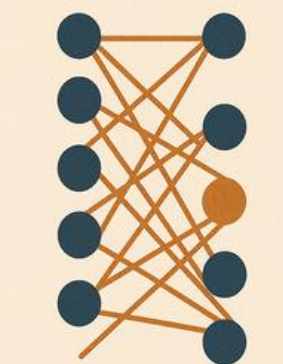
90s HIP-HOP



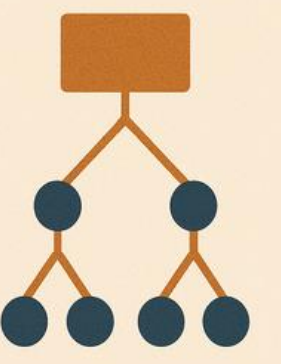
About Features

Feature 1		
Feature 2		
Feature 3		
Feature 4		
Feature 5		
Feature 6		
Feature 8		

NN vs Classsic ML



Neural Network



Classic Machine Learning

What is MIR & DSP?

MIR

- Music Information Retrieval (음악 정보 검색)
- Music Recommendation
- 음악적 특징을 뽑아 어떤 곡인지 판단

DSP

- Digital Signal Processing (디지털 신호 처리)
- FFT & STFT
- 음악 신호를 수학적으로 변환



90s Hip-Hop

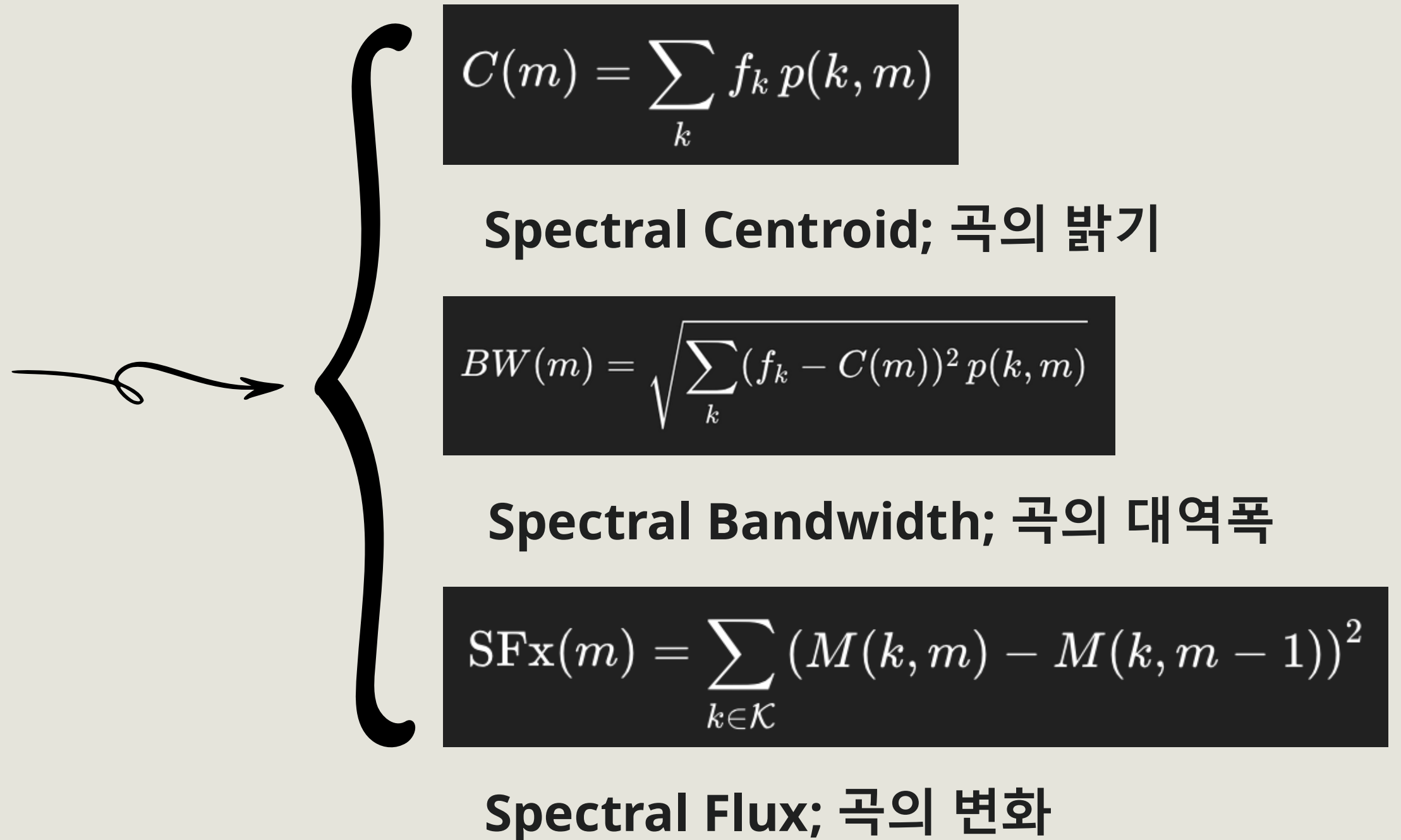


About Features

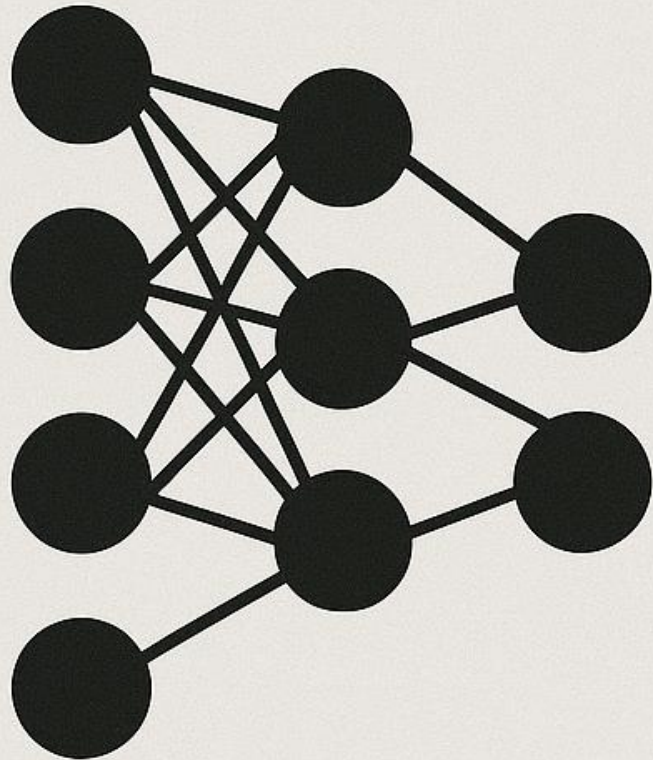
$$X(k, m) = \sum_n x[n] w[n - mR] e^{-j2\pi kn/N}$$

$$f_k = \frac{kF_s}{N} \quad p(k, m) = \frac{|X(k, m)|^2}{\sum_j |X(j, m)|^2}$$

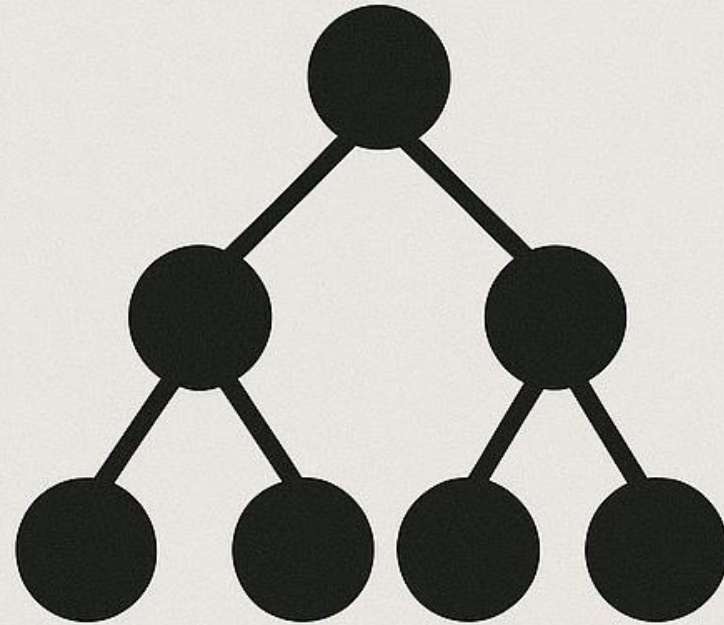
From FFT & STFT



NN



Classic ML (RF)



NN vs Classic ML

Neural Network

- Hidden Layer(은닉층)
- 복잡한 비선형 패턴 표현 가능
- 데이터가 충분히 많을 때 적합

Random Forest

- if-else 모음
- 비선형 패턴 표현도 가능하나 깊은 표현은 불가능
- 적당한 데이터 규모에서 적합

Conclusion & Future Work

Conclusion

- 8개 피쳐로 80% 수준의 분류
- Neural Network < Random Forest
- 스펙트럼 피쳐 설계로 장르 특성 해석 가능

Future Work

- 다른 장르로의 확장
- 피쳐 계산 수식 직접 설계
- 플레이리스트 추천 시스템으로의 응용

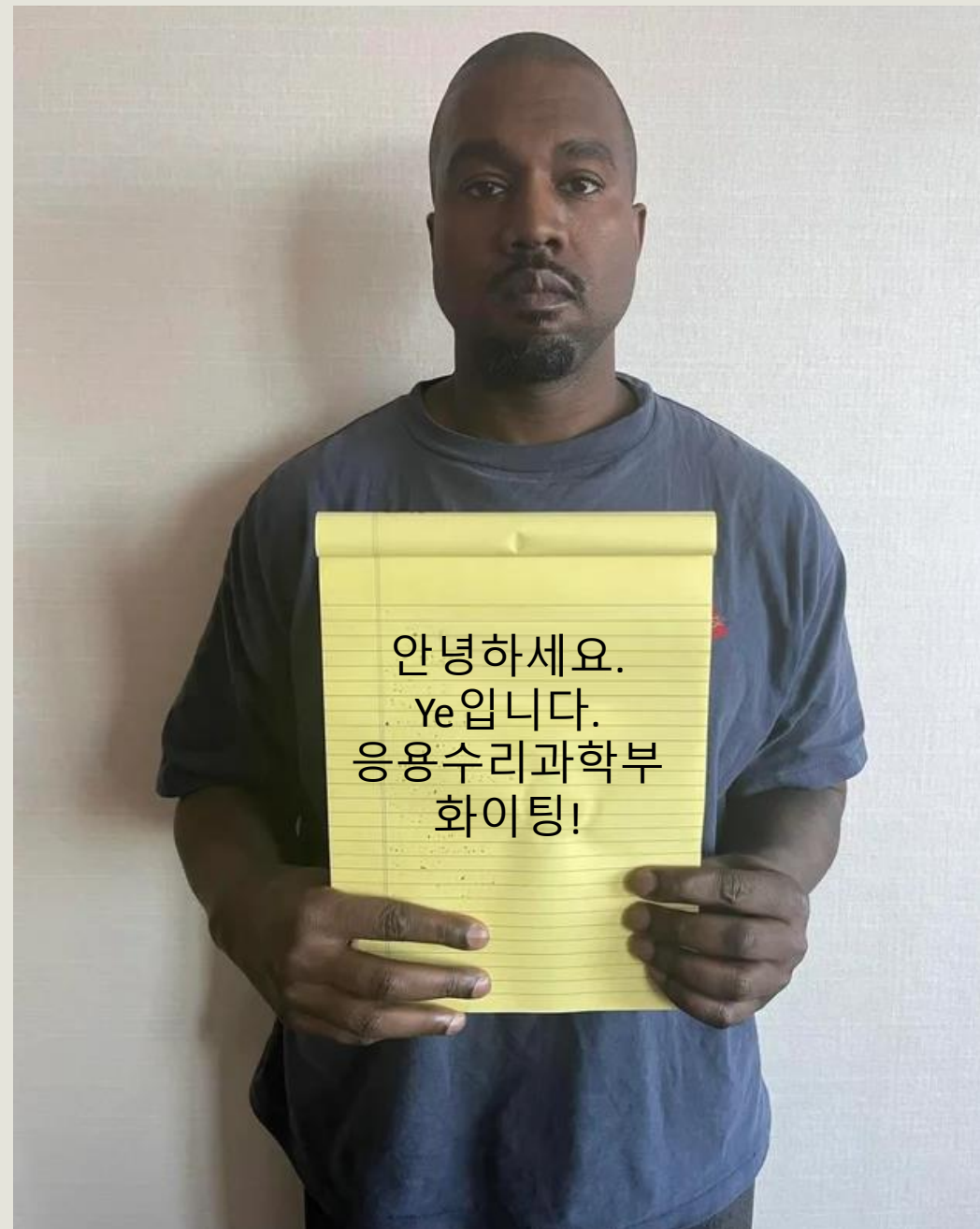
80%

NN < RF





Thank You For Listening!



안녕하세요.
ye입니다.
응용수리과학부
화이팅!